

LandWeb v3: Provenance

Input-data manifest, bibliography sync, and reproducibility (design in progress)

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⚠ Warning

Draft report: pending human review

Portions of this document were drafted with the help of generative AI and have not been fully reviewed. The input manifest, the bibliography sync, and the per-report build-provenance appendix are **implemented**; the **output manifest** and the **ODD / TRACE narratives** are designed here but not yet fully built (see the relevant sections).

1 Overview

This report documents how LandWeb v3 records the **provenance of both its inputs and its outputs** — where each dataset came from, and how each result can be traced back to the data + code + environment that produced it. **Inputs and outputs are documented together here** (rather than in separate reports): they are the two halves of one end-to-end lineage, and a reader auditing a result needs both in one place. The design has five parts, at different stages of maturity:

1. an **input-data manifest** (implemented) — one machine-readable record per input dataset (below);
2. an **auto-synced bibliography** (implemented) — data-source citations derived from the manifest;
3. a **per-report build-provenance appendix** (implemented) — repo commit, pinned module/package revisions, and software environment stamped into every rendered report (see the appendix to this document, and the shared `_provenance-appendix.qmd` include);
4. **output provenance and lineage** (design below; partly implemented) — each output artifact tied to its producing `targets` target, the module revisions that generated it, its input lineage, and the run parameters (study area, replicates, seed); and
5. **reproducibility-standards alignment** (below) — how the above maps onto FAIR, PERFICT, the Ten Simple Rules, ODD, and TRACE.

2 Input-data manifest

The manifest (`../_input_manifest.json`) is a schema.org `DataCatalog` (JSON-LD) with one record per input dataset, built with the `workflowtools` manifest API (`input_manifest_record()` / `register_input()`) from `R/build_input_manifest.R`. Each record carries a stable `id`, the source (type + URL), a `retrieved_at` timestamp, the local path, version/vintage, license, and a citation pointer. It currently registers **11** datasets:

Table 1: Registered input datasets (from `_input_manifest.json`). Each record additionally stores its `retrieved_at` timestamp, local path, license, and citation pointer.

ID	Dataset	Source	Version / vintage
current-condition-v2	Current-condition layers, v2 era (CASFRI / Pickell / AVI-derived)	drive	v2 era
gadm-canada-provinces	GADM Canada provincial/territorial boundaries (level 1)	http_download	GADM 4.1
knn-beaudoin-2001	Beaudoin kNN forest attributes (2001 base year)	http_download	2001 base year, v1 (250 m)
lthfc-sls-v3	Spray Lake Sawmills fire-cycle variant (SLS v3)	drive	v3 (NAD83 / UTM 11N; field LTHFC)
lthfc-nwab-aoi	Northwest Alberta area-of-interest outline	manual_drop	2026
lthfc-v10	LandWeb long-term historic fire-cycle map, v10 (2026)	drive	v10 (2026; NW Alberta = NWAB Intermediate)
lthfc-v8c	LandWeb long-term historic fire-cycle map, v8c (previous)	drive	v8c (NAD83 / UTM 11N; field LTHFC)
nwab-data-report-2026	Northwest Alberta NRV – Data Report (June 2026)	literature	June 2026
nwab-final-report-2026	Northwest Alberta NRV – Final Report (June 2026)	literature	June 2026
scanfi-species-2020	SCANFI v2 per-species crown closure (2020)	drive	2020, v2 (stamp 20260119)
scanfi-structure-2020	SCANFI v2 stand age, biomass, land cover (2020)	drive	2020, v2 (stamp 20260119)

The manifest is hand-curated (run `R/build_input_manifest.R` to regenerate) and committed to the repo, so it travels with the code. It deliberately omits content hashes for the multi-GB rasters; `sha256` can be added per record where a hash is worth carrying.

3 Bibliography auto-sync

`workflowtools::sync_manifest_to_bibtex()` reads the manifest and writes a sidecar `citations/data-sources.bib`, so that data-source citations do not have to be hand-maintained alongside the methods references. It is wired as the `data_sources_bib` pipeline target and deduplicates against the hand-maintained `citations/references.bib`.

Adopted convention. Data sources that are *also* cited as methods (SCANFI, Beaudoin *k*NN, the NWAB reports) keep a single hand-maintained entry in `references.bib` and are marked `external` in the manifest (so they are not duplicated into `data-sources.bib`). Any data source that needs a citation but does *not* already have one in `references.bib` carries its BibTeX in the manifest record and is emitted to `data-sources.bib` automatically. Reports list both `.bib` files in their front matter, so either path resolves. This keeps the methods bibliography curated while making data-only citations a by-product of registering the dataset — no separate hand-maintenance.

4 Per-report build-provenance appendix

Every report ends with a **Build provenance** appendix (see the appendix to this document, below), generated at render time from `workflowtools`' `info_*`() family. It is a shared Quarto include — `reports/_provenance-appendix.qmd` — that any report adopts with a single line (`{{< include _provenance-appendix.qmd >}}`), so all reports are stamped identically. It records:

- **build identity** — repository branch, HEAD commit, and render timestamp (`info_git()`);
- **pipeline state** — the pinned SHA of every SpaDES module and owned package at render time (`info_submodules()`), the same revisions report 03 documents;
- **software environment** — R version, operating system, and GEOS / GDAL / PROJ versions (`info_session()` / `info_spatial_libs()`).

For the simulation-run report (05) it will additionally carry the **base seed and per-replicate seed derivation**; those apply once the main simulation is wired. (Note: `workflowtools`' `reproducibility_receipt()` emits HTML `<details>` blocks aimed at HTML output, so the PDF reports format the same `info_*`() data into tables directly.)

This appendix is what makes the v2-vs-v3 output comparison (report 04) interpretable: it pins each PDF to the code and environment that produced it, so a reader can attribute an output difference to code vs. inputs vs. randomness.

5 Output provenance and lineage

Output provenance is the mirror of the input manifest: every artifact the pipeline produces — a summary CSV, a map, a rendered figure, a simulation raster — should be traceable to **what produced it**. The design has four ties per artifact:

1. **Producing target** — the `targets` node and its command that wrote the file;
2. **Code revision** — the SpaDES module / package SHAs in effect at build (the same `info_submodules()` set stamped in the build-provenance appendix);
3. **Input lineage** — which registered inputs (`_input_manifest.json` ids) fed the target, transitively; and
4. **Run parameters** — study area, replicate count, and random seed.

Much of this is **already captured, not yet surfaced**:

- **targets records it natively.** `targets::tar_meta()` stores, per target, the command, the upstream dependencies, a content hash, the seed, and the build timestamp — i.e. the store *is* a content-addressed provenance graph. `targets::tar_network()` exposes the dependency edges (the input-to-output lineage) directly.

- **SpaDES.targets** records **per-run** **saves**. Each simulation stage returns an `outputs(sim)` manifest of the files it wrote (per-timestep saves, module `registerOutputs()` dumps, `Plots()` figures), discovered dynamically rather than declared a priori, plus a companion `format = "file"` target that hashes every saved file.

What remains (planned): a small producer that reduces `tar_meta()` + the run configuration into an **output manifest** in the same shape as the input manifest (one record per artifact: id, producing target, code SHA, input ids, run parameters, hash), and wiring the **build-provenance appendix of a simulation report (05) to filter that manifest to the artifacts that report used** — closing the loop from a figure in a PDF back to the exact inputs, code, and seed behind it. This is scoped to land with the main-simulation and summary stages (reports 04/05).

6 Reproducibility standards

LandWeb v3 targets a **PERFICT**-style workflow (McIntire et al. 2022; Barros et al. 2023) — frequent **P**redictions and **E**valuations of **R**eusable, **F**reely-accessible, **I**nteroperable models, built in **C**ontinuous, **T**ested workflows — and the manifest, appendix, and **targets** graph are the runtime evidence for that claim. The table below maps the provenance design onto the standards the project aims to meet; the “status” column is an honest self-assessment for the co-authors to refine.

Standard	What it asks for	LandWeb v3 status
FAIR (Wilkinson et al. 2016)	Findable, Accessible, Interoperable, Reusable data + workflows	Inputs are registered with source, license, and citation (manifest); outputs FAIR-ify once the output manifest lands. Access to some inputs is controlled (proprietary/Drive).
PERFICT (McIntire et al. 2022; Barros et al. 2023)	Reusable, tested, continuous predictive-ecology workflows	Core aim of the v3 rebuild: targets continuous workflow, owned CI-tested packages, per-FMA reuse.
Ten Simple Rules (Sandve et al. 2013)	Version control, recorded seeds, tracked intermediate steps, avoid manual steps	Met by targets (recorded seeds/hashes, no manual steps) + pinned revisions; NEWS now documents module history (report 03).

Standard	What it asks for	LandWeb v3 status
ODD (Grimm et al. 2006)	A standard narrative description of the model	Partly served by the manual (#128) + report 00; a formal ODD write-up is outstanding.
TRACE (Grimm et al. 2014)	Documented model development, testing, and analysis	Partly served by reports 01-03 (data, methods, changes) + module tests; a consolidated TRACE document is outstanding.

The manifest + build-provenance appendix are the *runtime evidence* an ODD/TRACE narrative would point to. Fleshing those narratives out (and the FLM-calibration best-practices alignment, (Reese et al. 2024), report 07) is the next step for the co-authors.

References

- Barros, Ceres, Yong Luo, Alex M Chubaty, et al. 2023. “Empowering Ecological Modellers with a PERFICT Workflow: Seamlessly Linking Data, Parameterisation, Prediction, Validation and Visualisation.” *Methods in Ecology and Evolution* 14: 173–88. <https://doi.org/10.1111/2041-210X.14034>.
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- Reese, Gordon C, Brian R. Sturtevant, Caren C. Dymond, et al. 2024. “Best Practices for Calibration of Forest Landscape Models Using Fine-Scaled Reference Information.” *Canadian Journal of Forest Research*, ahead of print. <https://doi.org/10.1139/cjfr-2024-0085>.
- Sandve, Geir Kjetil, Anton Nekrutenko, James Taylor, and Eivind Hovig. 2013. “Ten Simple Rules for Reproducible Computational Research.” *PLoS Computational Biology* 9: e1003285. <https://doi.org/10.1371/journal.pcbi.1003285>.
- Wilkinson, Mark D., Michel Dumontier, IJsbrand Jan Aalbersberg, et al. 2016. “The FAIR Guiding Principles for Scientific Data Management and Stewardship.” *Scientific Data* 3: 160018. <https://doi.org/10.1038/sdata.2016.18>.

Provenance

- **Manifest:** `../_input_manifest.json`, built by `R/build_input_manifest.R` (hand-curated; `workflowtools::input_manifest_record()`), 11 records.
- **Bibliography sync:** `citations/data-sources.bib`, generated by `workflowtools::sync_manifest_to_` (the `data_sources_bib` target).
- **Method:** this report reads the committed manifest directly; rendered by the gated `landweb.extended_analyses` factory as a `tar_quarto()` target on the control node.

Appendix: Build provenance

This appendix pins the rendered PDF to the code, environment, and revisions that produced it, so a result can be reproduced or a difference attributed to code vs. inputs vs. randomness. It is generated at render time from the `workflowtools info_*`() family.

Build identity. LandWeb repository, branch `targets`, commit `ad683998`, rendered 2026-07-03 10:18 MDT with R version 4.6.1 (2026-06-24) on Ubuntu 26.04 LTS (GEOS 3.14.1, GDAL 3.12.2, PROJ 9.7.1).

Table 2: SpaDES modules and owned packages at their pipeline-pinned revisions when this report was rendered (`workflowtools::info_submodules()`).

Component	Kind	Pinned SHA
Biomass_borealDataPrep	module	f570094ea9
Biomass_core	module	32a3790870
Biomass_regeneration	module	b739da2d32
Biomass_speciesData	module	101ad29571
Biomass_speciesFactorial	module	01d4559d15
Biomass_speciesParameters	module	1e381a7881
HSI_Caribou_MB	module	1180d61743
LandMine	module	d7ccb083e1
LandWeb_output	module	af78cee648
LandWeb_preamble	module	65dc365f37
LandWeb_summary	module	99e229e3ac
NRV_summary	module	ca7b444fa2
burnSummaries	module	add6cc0bb4
timeSinceFire	module	30d242ea7b
LandWebUtils	package	cf29ea06e4
SpaDES.targets	package	b1f68a5467
crew.ssh	package	79c368f0fa
nrvttools	package	f9665fd7b0
spatialutils	package	a891aef9d5
workflowtools	package	561def4389

The full `sessionInfo()` and the exact `renv.lock` package pins accompany the run; input datasets are recorded in `_input_manifest.json` (report 06).